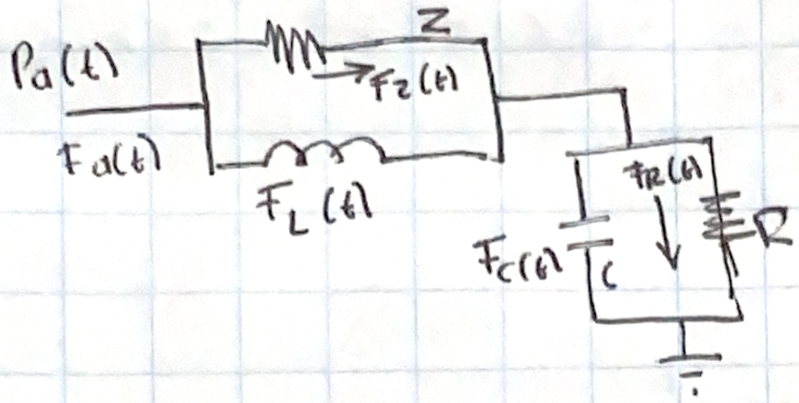
Análisis matemático



Función de transferencia

Ecuaciones principales

$$F_a(t) = F_z(t) + F_L(t) = F_c(t) + F_R(t)$$

$$F_z(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]\, dt$$

$$F_c(t) = C\, d\frac{P_p(t)}{dt}$$

$$F_R(t) = \frac{P_p(t)}{R}$$

Transformada de La place

$$F_a(s) = F_z(s) + F_L(s) = F_c(s) + F_R(s)$$

$$F_z(s) = \frac{P_a(s) - P_p(s)}{Z}$$

$$F_L(s) = \frac{P_a(s) - P_p(s)}{Ls}$$

$$F_c(s) = C_s P_p(s)$$

$$F_R(s) = \frac{P_p(s)}{R}$$

Procedimiento algebraico

$$\frac{P_a(s) - P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = C_s P_p(s) + \frac{P_p(s)}{R}$$

$$\left(\frac{1}{2} + \frac{1}{Ls}\right) P_a(s) = \left(Cs + \frac{1}{R} + \frac{1}{Z} + \frac{1}{Ls}\right) P_p(s)$$

$$\left(\frac{Ls + Z}{2Ls}\right) P_a(s) = \left[\frac{CRZLs^2 + 2Ls + RLs + RLs + RZ}{RZLs}\right] P_R(s)$$

$$\frac{P_0(s)}{P_a(s)} = \frac{\left[\frac{Ls+z}{2Ls}\right]}{\left[\frac{CRzLs^2+2Ls+RLs+R_2}{R2Ls}\right]} = \frac{RLs+Rz}{CRzLs^2+L(z+R)s+Rz}$$

Estabilidad en lazo abierto

- Normotenso

$z = 0.01$
$c = 1.5$
$R = 1$
$L = 0.001$

$$\lambda = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$a = (0.01)(1.5)(1)(0.001) = 1.5E^{-5}$
$b = (0.001)(0.01+1) = 1.01E^{-3}$
$c = (1)(0.01) = 0.01$

$\lambda_1 = -12.061$, $\lambda_2 = -55.271$

Presenta una respuesta estable y sobreamortiguada a un escalón unitario de entrada

- Hipotenso

$z = 0.01$
$c = 3.5$
$R = 1$
$L = 0.001$

$a = (0.01)(3.5)(1)(0.01) = 3.5E^{-5}$
$b = 1.01E^{-3}$
$c = 0.01$

$\lambda_1 = -14.428 + 8.805j$, $\lambda_2 = -14.428 - 8.805j$

Presenta una respuesta estable y subamortiguada a un escalón unitario

- Hipertenso

$z = 0.01$
$c = 1$
$R = 1$
$L = 0.01$

$a = (0.01)(1)(1)(0.01) = 1E^{-5}$
$b = 1.01E^{-3}$
$c = 0.01$

$\lambda_1 = -11.126$, $\lambda_2 = -89.873$

Presenta una respuesta estable y sobreamortiguada a un escalón unitario

Estrella

Error en estado estacionario

$$e(s)=\lim_{s\to 0} s\,P_a(s)\left(1-\frac{P_p(s)}{P_a(s)}\right)=e(s)=\lim_{s\to 0}\frac{1}{s}\left(1-\frac{LRs+RZ}{CLRZs^2+L[R+Z]s+RZ}\right)$$

$=0V$

$$\boxed{e(s)=0V}$$

Modelo de ecuaciones integrodiferenciales

$$F_Z(t)+F_L(t)=F_C(t)+F_R(t)$$

$$\frac{P_a(t)-P_p(t)}{Z}+\frac{1}{L}\int[P_a(t)-P_p(t)]\,dt=C\frac{dP_p(t)}{dt}+\frac{P_p(t)}{R}$$

$$\left(\frac{R+Z}{ZR}\right)P_p(t)=\left[\frac{P_a(t)}{Z}+\frac{1}{L}\int[P_a(t)-P_p(t)]\,dt-\frac{C\,dP_p(t)}{dt}\right]$$

$$P_p(t)=\left[\frac{P_a(t)}{Z}+\frac{1}{L}\int[P_a(t)-P_p(t)]\,dt-\frac{C\,dP_p(t)}{dt}\right]\frac{ZR}{Z+R}$$

Estrella